

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham

Dec 2025 – Feb 2026

Product Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer

Remote

- Architected a resilient **Puppeteer-based Node.js automation engine** for MCA company registrations, reducing manual filing time by **85%** via intelligent network-idle monitoring, **Vision API CAPTCHA** solving, and making good use of open API endpoints on the site.
- Built a **Go notification service** with a unified API for WhatsApp/email channels, provider-agnostic MSG91 integration, strict request validation, and Controller-Service-Provider separation.
- Designed scalable, production-ready infrastructure by **containerizing with Docker** and deploying to **Google Cloud Run**.

Google Summer of Code

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Developed a tool to segment **95 subparts of the brain** from MRI images using **deep learning models** with support for both **PyTorch** and **TinyGrad**.
- Optimized processing time by **87%** by utilizing **asynchronous functions** and **parallel processing**.
- Enhanced the **3D reconstructed surface visualization** system.

Open Source

Contributions

- **Invesalious**: shipped GSoC brain-subpart segmentation for **95 MRI regions** and supporting medical-imaging fixes across segmentation progress UI, wxPython compatibility, preferences loading, and Zip Slip path validation.
- **Kuadrant**: implemented MCP Gateway backend-outage grace handling, retaining advertised tools through transient upstream failures with `--remove-tools-grace-period`, status updates, and recovery/removal test coverage.
- **OPA**: updated `opa fmt` to preserve multiline single-entry arrays/objects instead of collapsing deeply nested Rego structures, with v0/v1 formatter golden-test updates.
- Contributed to IOOS, VideoLAN, CRIU, and Tiled.

Projects

Chess Engine

GitHub

- Built a **bitboard-native chess engine** in Go with legal move generation, king-safety filtering, make/undo, minimax, alpha-beta pruning, move ordering, quiescence search, and a transposition table.
- Implemented a basic **UCI command loop** and iterative deepening with timer controls so the engine can run from the CLI or under chess GUI protocol flows.

Murdoc

GitHub

- Built a self-hosted **AI security gateway** for LLM, HTTP tool, and MCP traffic, blocking **90%+** of prompt-injection and unsafe-tool attempts across a **150-case attack corpus** with false positives under **5%**.
- Added route profiles, RBAC runtime settings, PII redaction, allow/block rules, **OpenTelemetry** metrics/traces, health checks, and audit ledgers for **100%** of policy decisions.

Technical Skills

Languages: Python, Go, TypeScript

Frameworks: FastAPI, Echo, Node.js, Express, AWS

Tools: PostgreSQL, Redis, RabbitMQ, Docker, Prometheus, Go-Rod, Puppeteer